Reclaiming the Bus Stop

Illegal parking in bus laybys: the cost to On-Time Performance, a critical passenger-safety exposure — and a self-funding enforcement programme to fix both

Public Transport Agency · Bus Performance Study · July 2026

Executive summary

THE PROBLEM

346,000

bus arrivals obstructed by illegally parked vehicles every year — at 912 stops carrying 76% of peak arrivals

−0.85 pp

peak On-Time Performance lost network-wide (60 s per obstruction scenario)

2,900 h

of direct bus delay per year — measured from AVL data and an 11-stop field survey

THE FIX PAYS FOR ITSELF

AED 46M

year-1 fine revenue from camera enforcement (AED 200/violation, 56% ticket-to-cash)

+87M

5-yr NPV @7% — hybrid: fixed cameras at 512 stops + 20 portable units

< 1 year

payback on every element deployed; revenue declines only because deterrence works

CRITICAL SAFETY ISSUE — when the layby is blocked, passengers board and alight in the live traffic lane (field photo). Every one of the ~346,000 obstructed arrivals is a potential live-lane boarding. This is the risk that cannot wait for a business case.

CRITICAL SAFETY ISSUE

Passengers step off into moving traffic

346,000 / year

obstructed arrivals — each one a potential boarding or alighting in the live lane

The most vulnerable pay first

elderly passengers, wheelchair users, parents with children — level kerb access is exactly what the layby exists to provide

Zero-distance near-misses

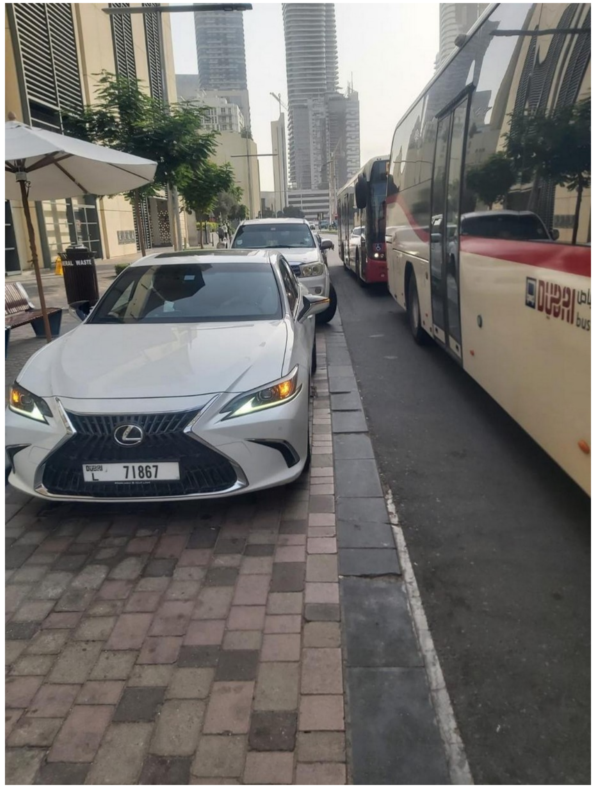
the door opens into the same lane cars use to overtake the stopped bus

A liability the Agency carries

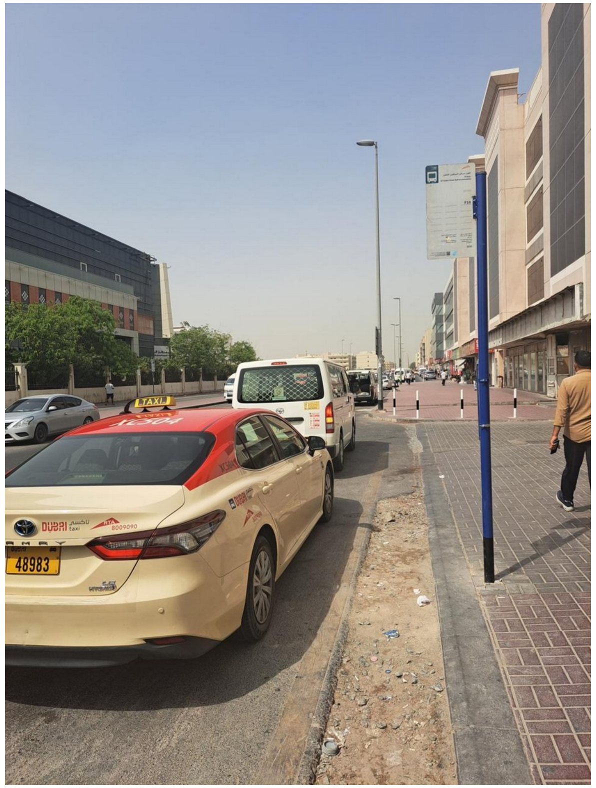
every peer city (NYC, London, Singapore) criminalised bus-stop parking on safety grounds before revenue grounds

Field photo, 28 Jul 2026 — layby blocked by a private car; the bus serves the stop from the carriageway and the passenger alights on the road, not the kerb.

Documented daily, across the network



Private car holds the layby — the bus loads in the traffic lane beside it



Taxi and van parked hard against the marked, signed bus stop (route F16)

Vans occupy the stop — the Al Quoz New Housing bus is forced to serve from the lane

Who blocks the stops

Taxis & limousines
labour/shuttle vans
private cars

60 cases

in one 3-hour morning at just 11 stops

All photos: PTA field observations, Jul 2026.

The evidence: one morning, 11 stops

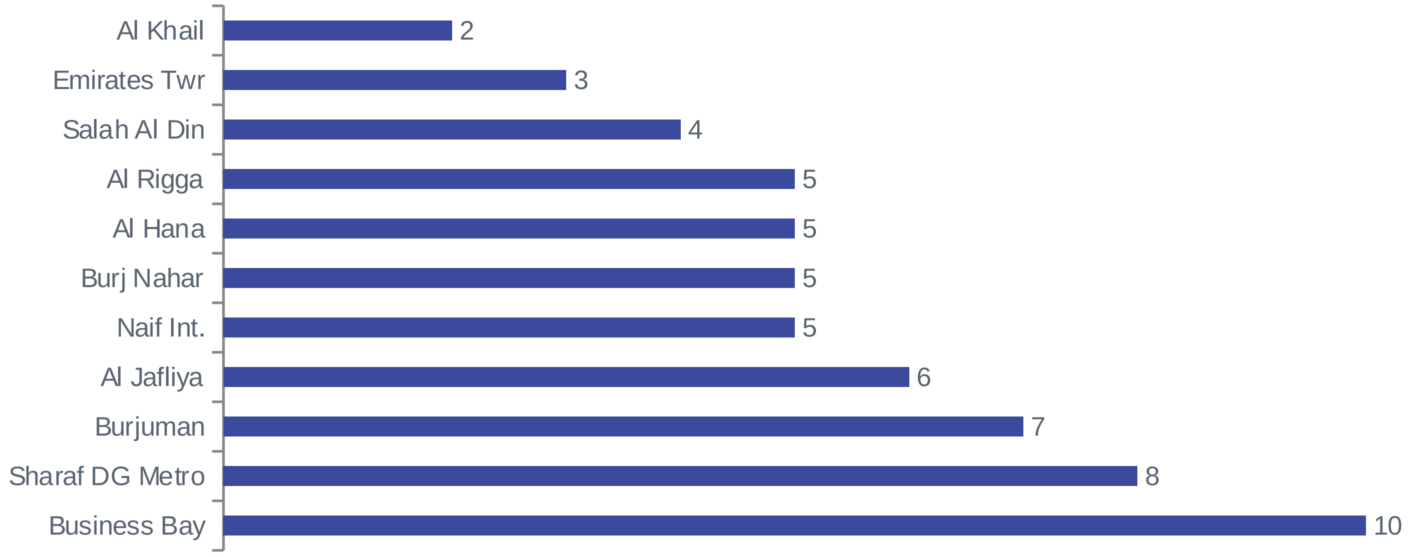
60
blocked-layby cases in 3 hours (07:00–10:00, one day)

1 in 21
AM-peak bus arrivals met an obstructed layby

87.2%
baseline OTP that day (38 routes, 72,679 stop events)

78.7%
OTP at Sharaf DG Metro — worst stop, most blockages (8)

Blocked-layby cases by location (07:00–10:00, single day)



Cross-check
AVL data (22 Jul 2026) confirms these stops carry 1,254 bus arrivals in the AM peak alone — and that late running already concentrates in peak hours.

Network-wide: 912 stops carry the risk

Every stop in the GTFS network was scored on the surveyed hotspot profile: peak bus throughput plus metro-station or mall adjacency.

Tier	Stops	Definition	Share of peak arrivals	Annual blockages
High	112	Matches surveyed profile (metro/mall + heavy throughput) — survey-calibrated rate	24%	≈ 182,000
Medium	400	Next 400 by peak throughput — 50% of survey rate (assumption)	36%	≈ 134,000
Low	400	Following 400 — 25% of survey rate (assumption)	16%	≈ 30,000
Total	**912**	**4 in 10 network stops**	**76%**	**≈ 346,000**

76%

of all peak bus arrivals happen at these 912 stops

7.5

violations per stop per day at High-tier stops

Biggest non-surveyed exposures: Ibn Battuta, Union Metro, Mall of the Emirates, Gold Souq, Dubai Mall Metro.

What it costs the network every year

346,000

bus arrivals obstructed per year
(912 stops, both peaks)

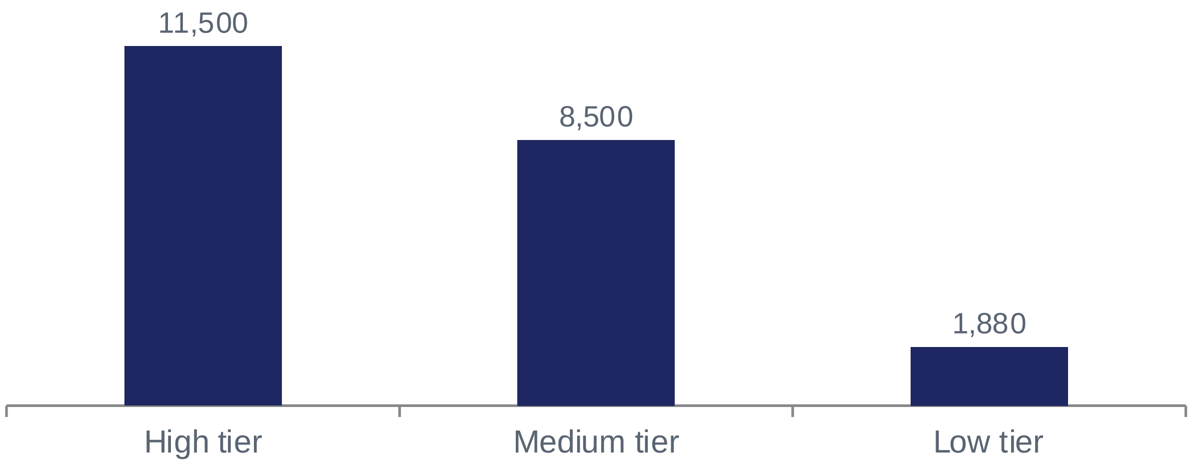
2,900 h

of direct bus delay per year
(at 30 s per obstruction)

−0.85 pp

peak OTP lost network-wide
(30–60 s scenarios: −0.35 to −0.85)

Extra late trips per year by tier (60 s per obstruction)



Why small delays matter

OTP counts a trip late beyond +5 minutes. The damage lands on trips already 4–5 minutes behind — 12.7% of peak stop events. A 30-second obstruction tips 1 in 38 of the trips it touches over the line; at 60 seconds, 1 in 16.

Enforcement pays: the violation-to-cash funnel

415,000
detectable violations / yr
cameras watch 24/7 — off-peak adds 20%

→

290,000
valid tickets (70%)
plate readable, stationary, de-duplicated

→

232,000
collected (80%)
at AED 200 per violation

→

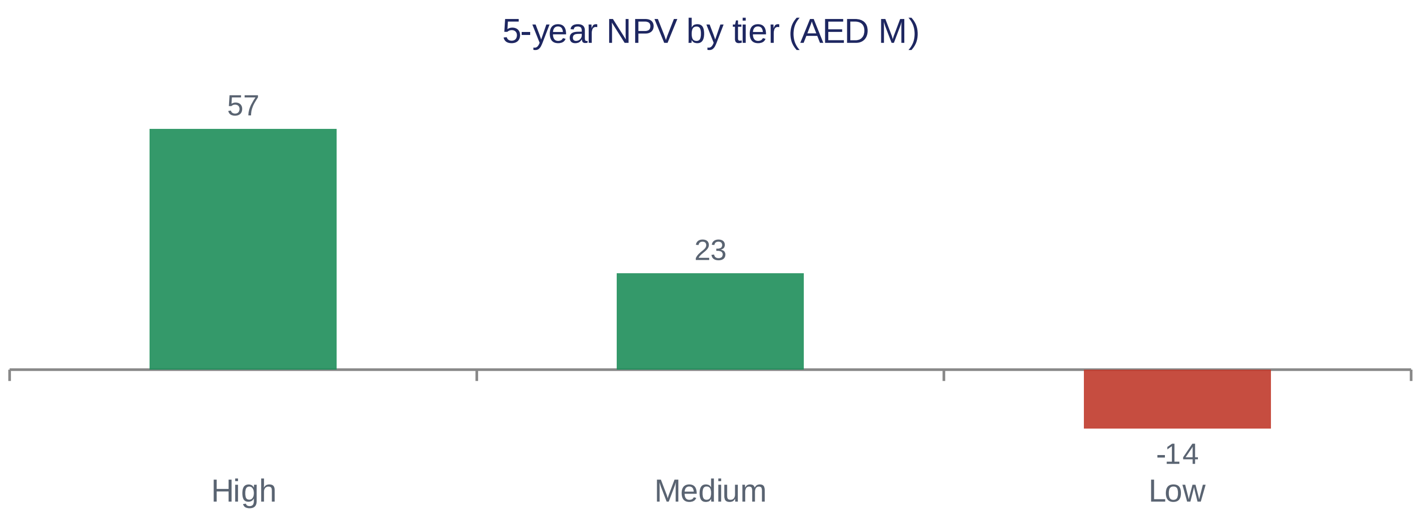
AED 46M
gross revenue, year 1
declining by design as deterrence works

Revenue declines 25%/yr — because deterrence is working

The honest pitch
Fines finance the system in years 1–3. The permanent return is OTP recovery and the end of live-lane boarding.

Option A — fixed cameras: invest where density pays

Tier	Capex	Revenue yr 1	5-yr NPV @7%	Payback
High — 112 stops	AED 4.5M	AED 24.4M	**+56.6M**	< 1 year
Medium — 400 stops	AED 16.0M	AED 18.0M	**+22.6M**	~1 year
Low — 400 stops	AED 16.0M	AED 4.0M	**−13.8M**	never



The verdict

Stop at High + Medium.

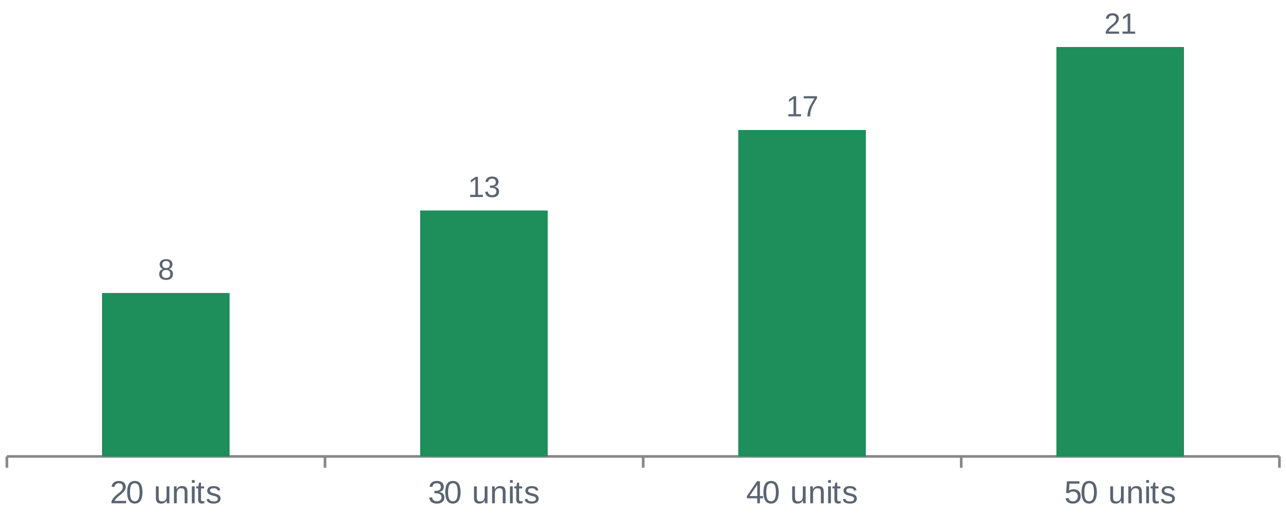
512 stops · AED 20.5M capex
5-yr NPV +79.2M · payback < 1 yr

The Low tier's violation density is too thin to carry a fixed installation — leave it to the portable fleet.

Assumed: AED 40k/site capex, 5k/yr opex (benchmark; pre-RFP).

Option B — portable cameras: small capex, follows the problem

5-yr NPV by fleet size, monthly rotation at High-tier stops (AED M)



7.5

violations seen per deployed camera per day

3.5×

daily revenue vs daily all-in cost per unit

Deployed at High-tier stops; deterrence decay 15%/yr under monthly rotation.

Economics per unit: ≈ AED 830/day collected vs ≈ AED 240/day cost (AED 15k capex + 60k/yr crew, vehicle, redeployment). Every fleet size is NPV-positive — value scales with units, and rotation adds a "cameras could be anywhere" deterrence halo that fixed sites lack. Trade-off: only 2–5% of stop-time covered, so OTP recovery is slower than fixed coverage.

The 5-year money story — hybrid programme

Fixed cameras at 512 High + Medium stops, plus 20 portable units. All figures AED, undiscounted cash per year.

PHASE 1 — deploy High tier + portable

PHASE 2 — extend to Medium tier

PHASE 3 — steady state, review Low tier

BREAK-EVEN ~ MONTH 7

Start	Year 1	Year 2	Year 3	Year 4	Year 5
−20.8M	+46.8M	+35.5M	+27.0M	+20.6M	+15.7M
capex: 512 fixed sites + 20 portable units	revenue opex −3.8M	revenue opex −3.8M	revenue opex −3.8M	revenue opex −3.8M	revenue opex −3.8M
	net +43.0M	net +31.8M	net +23.3M	net +16.8M	net +11.9M

+106M
cumulative cash by end of year 5

+87M
net present value @ 7% discount

+0.85 pp
peak OTP recovered — the return that compounds

Recommendation: the hybrid programme

Phase 1 · Months 0–6

Fixed cameras at 112 High-tier stops + 20 portable units.
Run the validation survey that firms Medium/Low rates.
Capex ≈ AED 4.8M.

Phase 2 · Months 6–18

Extend fixed coverage to the 400 Medium-tier stops.
Grow portable fleet to 30–50 as validated data directs.
Capex ≈ AED 16M — funded by Phase 1 revenue.

Phase 3 · Year 2+

Portable rotation covers the Low tier and emerging hotspots.
Integrate with the fines system; publish OTP recovery.
Review AED 200 vs AED 500 federal fine level.

+87M

5-yr NPV, hybrid programme (AED)

+0.85pp

peak OTP recoverable network-wide

< 1 yr

payback on every element deployed

Revenue is the financing bridge, not the goal: it declines as compliance improves, while the OTP and safety gains compound.

Every peer city has already moved

New York — MTA ACE

Bus-mounted cameras, escalating fines US$50–250

~5% faster buses, ~20% fewer collisions, ~30% fewer repeat violations

London — TfL

Bus stop clearways, CCTV-enforced 24/7, PCN £160

Near-universal kerb access; compliance underpins the iBus OTP regime

Singapore — LTA

Camera-equipped buses since 2008, ~S$150 fine

Sustained bus-lane and stop compliance

Sydney — TfNSW

Bus-zone offence ≈ A$390

Deterrence through penalty level

Dubai today: manual patrols only. This study models a conservative AED 200 fine — the federal schedule already allows AED 500 for stopping in a bus stop, which would scale every revenue figure ×2.5 and turn even the Low tier positive.

Decisions requested today

1. **Approve Phase 1**

   AED ~5M: fixed cameras at the 112 High-tier stops plus a 20-unit portable fleet. Self-funding within year 1.

2. **Confirm revenue retention**

   Agree with Dubai Police / DoF how bus-stop fine revenue is shared — the single biggest swing factor in the cash ROI.

3. **Commission the validation survey**

   1–2 weeks at ~20 Medium/Low-tier stops to replace the two assumed rate multipliers with measurements.

4. **Issue the enforcement RFP**

   ANPR fixed sites + portable units + fines-system integration, structured to allow bus-mounted cameras (NYC model) as a Phase 3 extension.

Full study, data and adjustable financial model: repo gtfslatest · STUDY.md · parking_otp_impact_model.xlsx